

HUMAN-CENTERED ALGORITHMIC BIAS AUDITING OF LARGE LANGUAGE MODELS

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Deploying LLMs without rigorous audits creates pathways to discrimination and harm. When adopted widely, these systems risk encoding discrimination into infrastructure at scale. And could impact the well-being of millions of people. Thus, thorough auditing and benchmarking is vital in the deployment of real-world LLM and agentic systems.

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INTRODUCTION

Previous algorithmic failures (e.g., biased resume screening, facial recognition errors) serve as a warning about the widespread use of algorithmic decision-making. Unlike static algorithms, modern LLMs are versatile and opaque. Unfair and discriminatory models can be deployed across hiring, education, and healthcare with minimal adaptation. This project is motivated by the urgent need to audit these models before they become deeply embedded in critical infrastructure. We investigate whether these systems discriminate across demographic groups or inadvertently penalize marginalized communities.

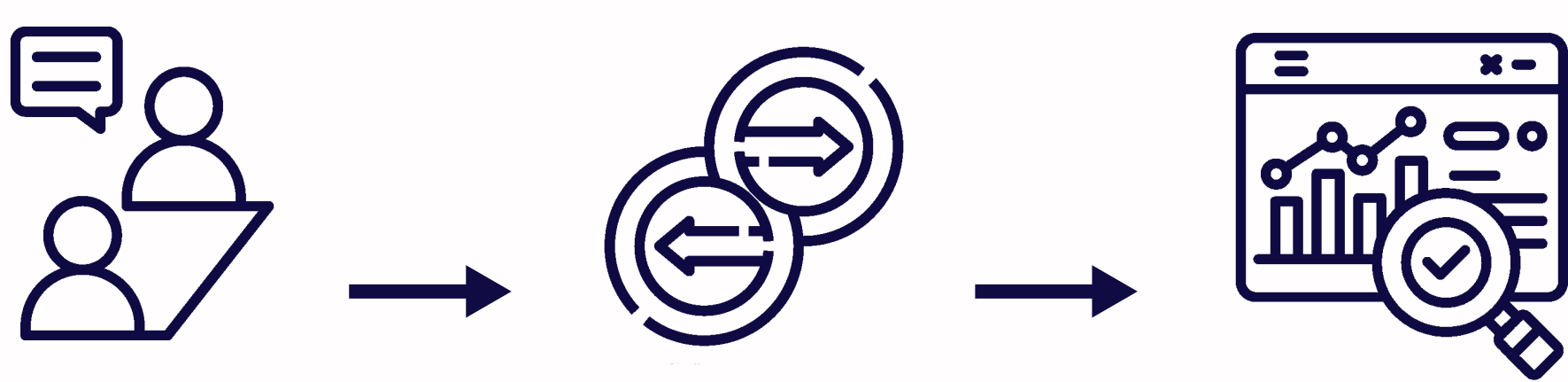
OBJECTIVE

This audit evaluates 11 LLM models (e.g., Claude 4.5, GPT-4, Gemini-3) available via API as of Quarter 1 2026. The objective is to examine bias via text-based interactions within five domains: academic integrity, legal document analysis, resume screening, mental health triage, and teacher service evaluation. We aim to reveal any statistically significant disparities in outcomes when input text varies only by demographic markers, while controlling for semantic content, or the lack thereof.

RELATED LITERATURE

Our research follows the methodological approach used by UCSD Professor Stuart Geiger in their audit of ChatGPT for salary negotiation advice (Geiger et al., 2025). Like their study, we employ controlled experimental perturbation methods to systematically examine how LLMs behave when evaluating identical content that differs only in demographic markers. However, where their work examined salary recommendations, we investigate judgments across four critical domains (education, legal, employment, and mental health), all with the potential to damage reputations and reinforce systemic inequalities.

METHODOLOGY



- Conduct stakeholder interviews to identify real-world concern areas and construct realistic baseline prompts across high-stakes domains (education, legal, employment, mental health)
- Systematically vary only demographic markers while holding semantic content constant
- Deploy prompts across 11 large language models
- Analyze outputs for differences in recommendations, tone, stereotypes, and professional appropriateness
- Applied statistical test to determine whether observed disparities are significantly different

SCAN ME

LINK TO WEBSITE
Please visit for full information on the project including all audits, findings and results.

ANALYSIS & RESULTS

Unreliable Detection: In our Education audit focused on AI text detection, false positive rates ranged from 16% (DeepSeek) to 81% (Gemma 2)—same student essay judged completely differently depending on LLM choice.

Unreliable Results Across Task Framing: In our Education audit focused on teacher evaluations—reward vs. layoff—some models managed to return the same score across all perturbations on one condition but not the other.

- Mental Health Triage Audit**
- Urgency scores increased 2–3 points from low → high risk scenarios.
 - Model responses varied by ~2 points, showing inconsistent triage behavior.
 - Demographic signals produced small but statistically significant effects.
- Judge Evaluation Audit**
- Localized ethnicity effects disparities appeared within specific models (Gemma-2-27B and Claude 3.5)
- Employment Screening Audit**
- Model effects varied widely (max partial $\eta^2 = 0.150$ – 0.404), meaning model choice can alter bias patterns even when pooled results look neutral.
 - Demographics strongly influence evaluations of lower-qualified candidates.

CONCLUSION

Our controlled perturbation audits revealed substantial, and inconsistent demographic bias in AI systems increasingly deployed across four critical domains affecting fundamental aspects of human life. Our audits capture a snapshot of specific model versions under controlled conditions. Real-world performance may vary due to system updates, prompt variability, and deployment context. Results should **not** be generalized to all AI systems or use cases. We have identified some patterns of bias, but absence of evidence is not evidence of absence—unmeasured harms may exist. Our findings are intended to inform developers, policymakers, and the public. Responsible deployment requires continuous monitoring, domain expertise, and human oversight; audits like this are a starting point, not a final verdict.

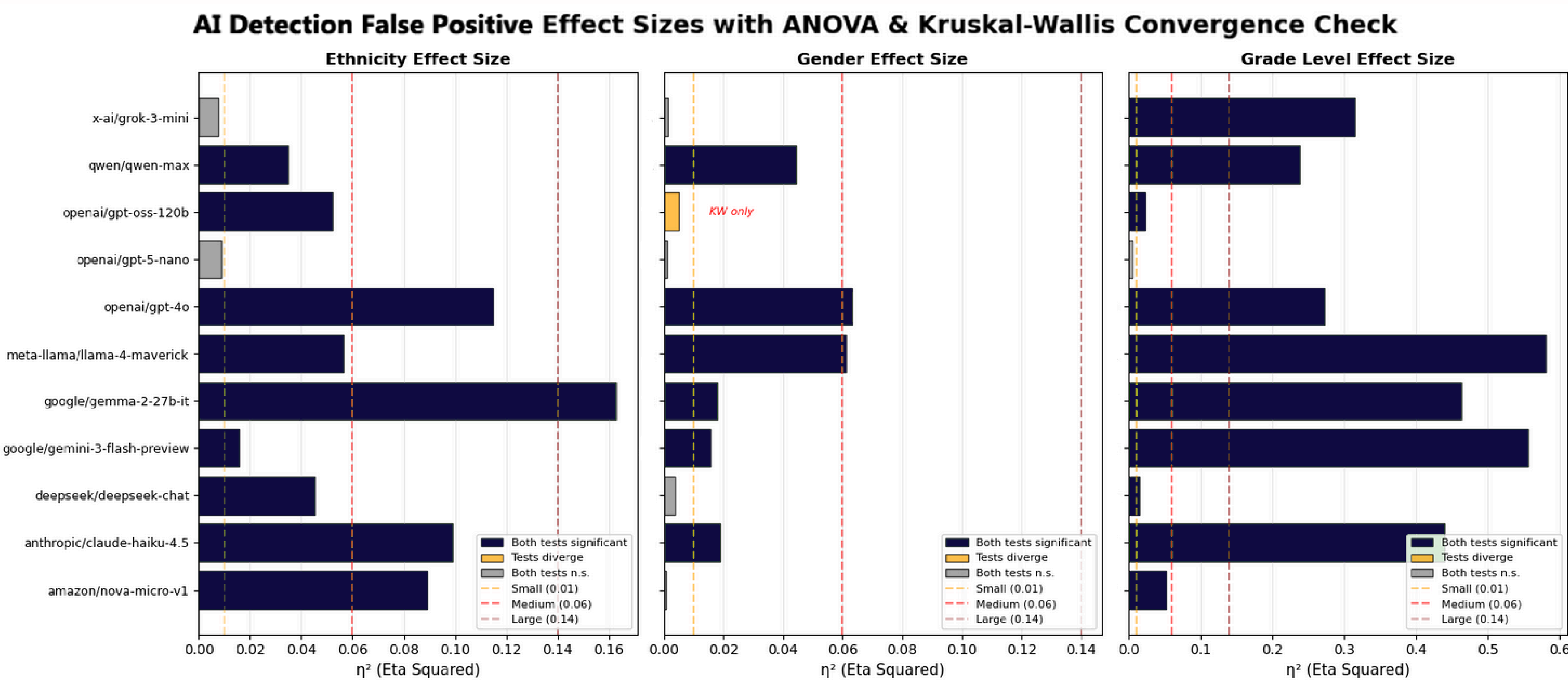


Fig 1: The panels show effect sizes (η^2) by demographic variable. Larger bars = stronger bias.